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PPS-reinforced poly(terphenylene) anion-exchange membranes with different quaternary ammonium groups for use in water electrolysers

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ABSTRACT

The nature, concentration and spatial distribution of quaternary ammonium (QA) groups is instrumental for the performance of anion exchange membranes, primarily by augmenting the ion exchange capacity (IEC). This study compares a series of poly(*meta*-terphenylene)-based ion exchange membranes (mTPN), which vary in the attached quaternary ammonium group. The IEC of the membranes is 1.97, 2.20 and 3.70 mmol OH^- g^{-1} for membranes functionalized with 1,4-diazabicyclo[2.2.2]octane (DABCO), trimethylamine (TMA), and methylated DABCO (DABCO-Me), respectively. The latter is obtained by methylation of the DABCO-functionalized membrane.

To control swelling, all membranes are reinforced with a poly(phenylene sulfide) (PPS) fiber mat. The highest conductivity, achieved by PPS/mTPN/DABCO-Me, is 140 mS cm^{-1} in 1 M KOH and 237 mS cm^{-1} in 4 M KOH. An alkaline stability test over 2 months revealed that membranes with TMA and DABCO have a stable in-plane conductivity, measured in pure water, while the membrane with DABCO-Me showed a strongly increased resistance. On the other hand, a 1 month stability test showed that the through-plane conductivity of the membrane with DABCO-Me in 1 M KOH hardly changed. This indicates that conductivity by absorbed KOH solution is more relevant than the conductivity provided by the quaternary ammonium groups, and promises stable operation in the electrolyser. This is supported by an electrolyser test, where a stable voltage of 1.9 V at 0.25 A cm^{-2} was achieved, with no observed change in membrane resistance over a test duration of 100 h (60 °C, 4 M KOH, PGM-free catalyst).

1. Introduction

Water electrolysis represents a crucial component of the clean energy transition. Currently, there are three main types of low temperature electrolysis technologies based on the use of different membranes: 1) alkaline water electrolysis (AWE), 2) proton exchange membrane water electrolysis (PEMWE), and 3) anion exchange membrane water electrolysis (AEMWE). AWE has been considered as one of the most mature and well-established technologies for water electrolysis. It has been widely employed for industrial hydrogen production, particularly in large-scale applications, due to its proven reliability and relatively low capital costs [1]. AWE typically uses a potassium hydroxide liquid electrolyte of high concentration, a porous diaphragm, and catalysts based on relatively abundant and cheap metals [2,3].

PEMWE offers the advantages of higher current density, higher differential pressures, quicker cold-start time, and a more rapid system response due to the application of a dense membrane instead of a porous diaphragm. However, the necessity for expensive catalysts based on

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iridium and platinum remains a drawback. AEMWE integrates the strengths of AWE and PEMWE by combining abundant catalyst metals like nickel, cobalt or iron-based materials with fluorine-free hydrocarbon-based membranes.

The nature, density and distribution of the AEM's quaternary ammonium groups strongly affect the stability and performance of the AEMs. Various quaternary ammonium groups have been utilized in AEMs, including groups such as trimethylammonium (TMA) [4], triethylammonium [5], and dipropylmethylammonium [6], benzimidazolium [7], imidazolium [8], pyridinium [9], guanidinium [10], methylpiperidinium [6] and diazabicyclooctane (DABCO) groups [11]. The influence of the quaternary ammonium group on the membrane properties is immense. It affects all important characteristics of the AEM, e.g. ionic conductivity, permselectivity, swelling, tensile strength, Young modulus and others [12–15]. From the above list, methylpiperidinium, TMA and DABCO appear to be the most alkaline stable systems [16].

Even more, warm water has a strong swelling and plasticizing effect on AEMs [17,18], and most commercial membranes recommended for use in water electrolysers are also available as a reinforced membrane [19]. Perfluorinated support materials based on extended PTFE have a high alkaline stability, but the high persistence of the fluorinated monomers, fluorinated processing agents and the polymer itself may soon result in a ban of these materials in areas where alternatives are available. Note that molecules with single CF_3 groups are expected to be still allowed. Fibermats made of polyphenylenesulfide (PPS), for example, have been investigated as diaphragm in AWE, and provide the required alkaline stability [20].

In this work, highly alkaline stable poly(*meta*-terphenylene)-based AEMs [21] are reinforced with a PPS fiber mat, and the effect of the quaternary ammonium group is investigated. A series of polymers is obtained by reacting the brominated precursor polymer (mTPBr) [22] with (a) trimethylamine (TMA) or (b) 1,4-diazabicyclo[2.2.2]octane (DABCO), and (c) methylating the DABCO-based polymer (DABCO-Me, Fig. 1), In this series, the IEC increases from 1.97 (DABCO) over 2.20 (TMA) to 3.70 mmol OH^-/g (DABCO-Me).

2. Experimental part

2.1. Materials

1,4-diazabicyclo[2.2.2]octane (DABCO; Tokyo Chemical Industry), trimethylamine (TMA, 30 wt% in water; Tokyo Chemical Industry), iodomethane (MeI; Daejung Chemicals), tetrahydrofuran (THF, >99 %, Daejung Chemicals), *N*-methyl pyrrolidone (NMP, Daejung Chemicals), sodium hydride (NaH, 60 % dispersion in mineral oil), potassium hydroxide (KOH flakes; Daejung Chemicals), potassium carbonate (K_2CO_3; Sigma Aldrich), sodium chloride (NaCl; Daejung Chemicals), potassium nitrate (KNO_3; Daejung Chemicals), ethanol (Daejung Chemicals) were all used as received. mTPBr was synthesized according to the literature [22]. PPS fiber mats (wet-laid veils, 70–90 μm thickness, areal weight: 10 g/m^2) were obtained from TFP (Technical Fibre Products, UK).

2.2. Preparation of reinforced membranes

2.2.1. TMA-based membranes

1 mmol of mTPBr was dissolved in 4.12 mL of THF. The solution was stirred overnight at room temperature until completely dissolved. The solution was cast on a glass plate with a doctor blade set to 100 μm. After the layer had completely dried, the mTPBr solution was cast again with a 200 μm blade. A 90 μm thick PPS mat (10 × 10 cm^2) was embedded on the wet film, and the solution was cast lastly on the PPS mat with the blade gap adjusted to 700 μm. Finally, the membrane was vacuum dried for 48 h at 120 °C. Then membranes were immersed in 30 ml of TMA solution at room temperature for quaternisation. After 7 days, the membranes were transferred to deionized water and stored overnight to remove any remaining TMA. The membranes were noted as PPS/mTPN/TMA.

2.2.2. DABCO-based membranes

1 mmol of mTPBr and 2 mmol of DABCO were dissolved in 3.556 mL of NMP. The solution was stirred overnight at 60 °C until completely dissolved. The solution was cast on a glass plate with a doctor blade set to 100 μm. After the layer had completely dried, the polymer solution was cast again with a 200 μm blade. A 90 μm thick PPS mat (10 × 10 cm^2) was embedded on the wet film, and the solution was cast lastly on the PPS mat with the blade gap adjusted to 700 μm. Finally, the membrane was vacuum dried for 24 h at 60 °C. Dry membrane was immersed at 80 °C in DI water, and afterwards dried under vacuum at 80 °C until constant weight was reached. The membranes were noted as PPS/mTPN/DABCO.

For methylation, PPS/mTPN/DABCO membranes were immersed in 8.5 wt% iodomethane in ethanol at room temperature. After 7 days, the membranes were moved to 1 M potassium carbonate solution and kept overnight to remove residual iodomethane. The methylated membranes were noted as PPS/mTPN/DABCO-Me.

2.3. Membrane characterization

2.3.1. Water uptake and swelling ratio

PPS/mTPN membranes with the size of 1 × 4 cm^2 were ion exchanged to their chloride form by immersing them in 1 M NaCl solution for 24 h. The membranes were moved to DI water. After 24 h of immersion at room temperature, wet weight and dimensions in thickness and length were measured, and noted as w_{wet}, t_{wet}, and l_{wet}, respectively. The samples were then vacuum-dried at 60 °C for 24 h to obtain the dry weight and dimensions in thickness and length, noted as w_{dry}, t_{dry}, and l_{dry}, respectively. Weight and thickness were measured by

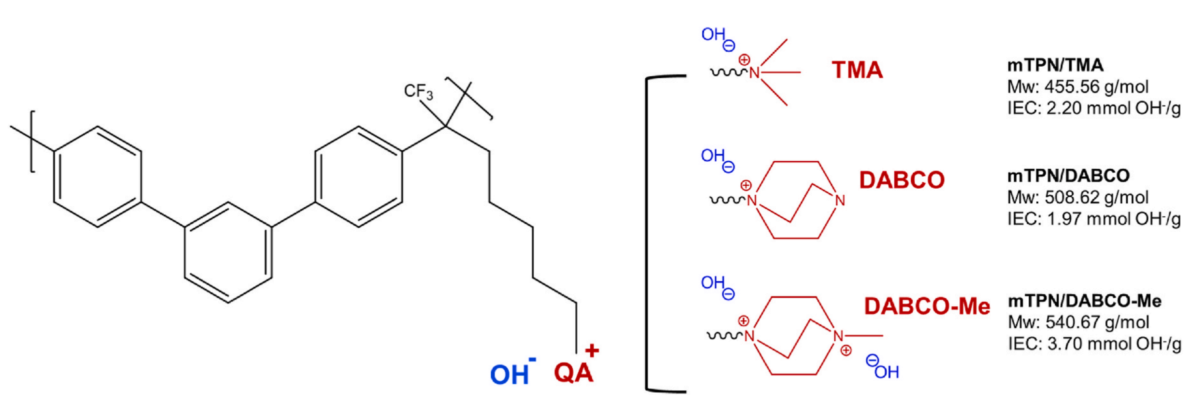

Fig. 1. Structures of the investigated polymers.

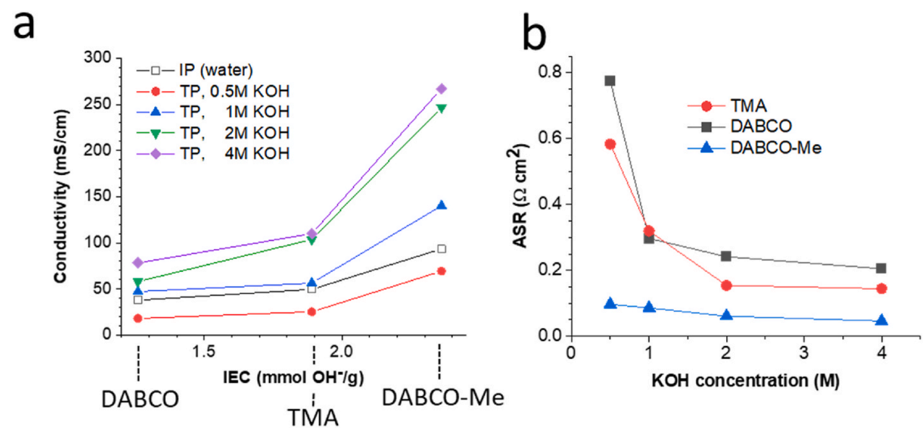

Fig. 2. (a) In-plane (IP) and through-plane (TP) conductivity and (b) area specific resistance (ASR) of hydroxide exchanged PPS/mTPN/x membranes at room temperature. The thickness in 2 M KOH solution was 140, 160 and 151 µm, respectively, for membranes modified with DABCO, TMA and DABCO-Me. IEC values used in Fig. 2a are based on the experimental values for PPS reinforced membranes (see Table S2 for experimental IEC values).

Ohaus balance and Mitutoyo thickness gauge 547-300S, respectively, while length was measured by taking a photo of the membranes and analysing the length using GIMP Software ver 2.10.28. From these wet and dry values, water uptake and swelling values were calculated.

$$\text{Water uptake }(\%) = \frac{W_{wet} - W_{dry}}{W_{dry}} \times 100\%$$

$$\text{Swelling ratio in thickness} \Big/ SR_T\ (\%) = \frac{t_{wet} - t_{dry}}{t_{dry}} \times 100\%$$

$$\text{Swelling ratio in length} \Big/ SRL\ (\%) = \frac{l_{wet} - l_{dry}}{l_{dry}} \times 100\%$$

$$\text{Swelling ratio in volume} \Big/ SR_V(\%) = \frac{((100 + SR_L)\ (100 + SR_L)\ (100 + SR_T))\text{-}1000000}{1000000} \times 100\%$$

2.3.2. Through-plane and in-plane conductivity

Through-plane conductivity at room temperature was obtained by measuring the membrane resistance using electrochemical impedance spectroscopy (Zive SP1, Wonatech). Equilibrated membranes (circle, diameter of 2.5 cm) which were previously immersed in KOH solution for 24 h were clamped between two gold-coated disk electrodes with an active area (*A*) of 1.767 cm^2 and measured in the frequency range of 1000 Hz to 0.5 MHz. Resistance of the membrane was measured for 1, 2, and 3 membrane layers, under the assumption that membrane/membrane interfacial resistance is negligible. The average thickness of the samples was noted as *t*. The resistance of the stacks was then plotted as a function of membrane thickness. A linear trend was observed and the slope ($\Omega\ cm^{-1}$) was used for the conductivity and area specific resistance (ASR) calculation by following equation.

$$\text{Through-plane conductivity }\left(\text{mS cm}^{-1}\right) = \frac{1000}{\text{slope}\cdot A}$$

$$\text{ASR }\left(\Omega\ \text{cm}^2\right) = Slope \cdot A \cdot t$$

For in-plane conductivity at room temperature, membranes (rectangular, 4 × 1 cm^2) were exchanged to carbonate form by immersing the membranes in 1 M K_2CO_3 for 24 h. Excess salt on the membrane was removed by immersing in DI water for another 24 h. Ion exchanged membranes were assembled into a Membrane Conductivity Cell (MCC) from WonATech-Zive Lab. The electrode distance was 1 cm. The cell was placed into a 1 L beaker which was filled up to three quarters full with pure water, and nitrogen was constantly bubbled through the water to exclude carbonates. The four electrodes of the MCC were connected to a Zive SP1 Electrochemical Workstation, and the electrochemical purging of carbonate ions from the membrane was accomplished by supplying a voltage of 2V for 30 min, further called as constant-mode measurement. After purging, current and voltage were measured by switching to cyclic mode by setting the voltage to 0, 0.1, and −0.1V as an initial, middle, and final voltage, respectively, further called as sweep-mode measurement. The measurement was performed in the sequence mode of constant and sweep for 18 h. Voltage value and obtained current value from sweep mode were plotted, and the slope was calculated as a resistance of the membrane (R_{mem}) in in-plane direction. After the measurement was done, the thickness of the now hydroxide exchanged membrane was measured (t), and the conductivity was calculated by following equation.

$$\text{In-plane conductivity }\left(\text{mS cm}^{-1}\right) = \frac{1000 \times \text{electrode distance}}{R_{mem} \times \text{membrane width} \times t}$$

2.3.3. Alkaline stability

Alkaline stability was measured by monitoring through-plane and in-plane conductivity of membrane samples. For through-plane conductivity, membranes were immersed in 1 M KOH at 60 °C in PTFE vials and the conductivity was measured after 1 and 30 days. The values were compared to day 0 where the membrane was immersed overnight in 1 M KOH at room temperature.

For in-plane conductivity, the membranes were immersed in 1 M KOH at 60 °C for 1, 3, 5, 7 days, then every week until 8 weeks. In every measurement, the membrane was moved to DI water for 5 min, then the measurement was conducted for 18 h. The conductivity was measured by taking the average of the value which was levelling off during the experiment. After measurement was done, the KOH solution was changed to the fresh one. The degraded membranes after 8 weeks of alkaline stability test were used for vapor permeation test and compared with the fresh membranes.

2.3.4. Additional characterisation methods

Details for measuring scanning electron microscopy (SEM), fourier-transform infrared spectroscopy (FTIR), ion exchange capacity (IEC) and water permeability are shown in supporting information.

2.4. Water electrolysis test

Commercially available cathode H2GEN-M and anode OXYGN-N catalysts offered by CENmat were used to prepare cathode and anode catalyst layers, respectively. Catalyst coated membrane (CCM) approach has been used for membrane electrode assembly (MEA) preparation. Catalyst load was 2.5 mg cm^{-2}. PSEBS-CM polymer binder was used as catalyst layer polymer binder [23]. Catalyst to polymer binder ratio was 93/7 wt%.

The catalyst ink was deposited on the dry PPS/mTPN/DABCO-Me membrane without any modification. Ultrasonic tip with diameter of 0.5 mm and ultrasound power of 6 W was used for deposition. The ink was stored in 10 cm^3 syringe and dosed to the tip by a hose by an automatic pump with flow of 0.5 cm^3 min^{-1}. During the deposition, nitrogen gas with flow of 50 cm^3 min^{-1} was used. In every step, 4 layers of catalyst ink were applied, then the CCM was let to dry for 5 min and weighted. Spraying was repeated until the required amount of catalyst in the layer was obtained.

After the catalyst layer deposition the CCM was immersed in 5 wt% DABCO ethanol solution for 24 h at room temperature to functionalize the binder in the catalyst layer. Then the sample was washed with deionized water and immersed in 1 M NaOH solution for 24 h to transfer the membrane to the hydroxide form.

For the alkaline water electrolysis tests, liquid KOH electrolytes of concentrations 0.1, 1 and 4 M KOH were used at two different temperatures, 50 and 60 °C, and fed to both electrodes. A detailed description of the electrolysis cell assembly and image of the individual parts are provided in SI (Fig. S1). Measurement covered short-term experiments by means of load curves and electrochemical impedance spectroscopy measurements (EIS). Load curves were measured in potentiostatic regime in the range of the cell voltage from 1.5 to 2.0 V with steps of 0.05 V. At each step, current passing through the cell was measured for 1 min. Stable values of the current were averaged and the result was plotted in the form of load curve. EIS was measured at cell voltages of 1.5, 1.6, 1.7 and 1.8 V in the frequency range from 20 kHz to 0.1 Hz. Maximum amplitude of the perturbing signal was set to 20 mV. Stability test was performed at constant current density of 250 mA cm^{-2} in the range of 100 h at 60 °C in 1 and 4 M KOH. Every 20 h, load curve and EIS measurements were performed in order to have more information on the electrolysis cell stability.

The equivalent electrical circuit shown in Fig. S5 was used to evaluate the measured EIS spectra. Rs (Ω) stands for cell ohmic resistance, Rp1 and 2 (Ω) represent polarization resistances and CPE1 and 2 are constant phase elements. Due to the setup of the electrolysis single cell, Rs value can be related mainly to the resistance of the used MEA. Rp values are connected with the charge transfer across the electrolyte|electrode boundary. Rp values are thus related to the kinetics of the electrode reactions on anode (oxygen evolution reaction) and cathode (hydrogen evolution reaction). However, it is not possible to unequivocally address Rp values to specific electrode so the Rp1 and Rp2 are used in this paper. CPE1 and 2 are used due to the presence of the charged double layers formed at the electrolyte|electrode boundaries.

3. Results and discussion

3.1. Membrane fabrication

The bromoalkylated precursor polymer mTPBr was synthesized according to the literature [22]. For preparation of TMA-based membranes, mTPBr was filled into the pores of a porous PPS fibermat, and the PPS/mTPBr membrane was reacted with TMA in a heterogeneous process by immersing it in an aqueous TMA solution. For preparation of DABCO-based membranes, mTPBr was dissolved in NMP and reacted with DABCO, before the solution was filled into the pores of PPS fibermats. The low viscosity of the pore-filling solution and the succesfull preparation of the pore-filled membranes indicates that crosslinking of mTPBr by reaction of DABCO with two polymer chains did not occurr or at least did not play a significant role. This allows to utilize the unreacted second amine group of DABCO. By immersing the PPS/mTPN/DABCO membranes in ethanolic 8.5 wt% iodomethane solution, the unreacted amine groups can be methylated (DABCO-Me, Fig. 1). To achieve quantitative methylation, the membranes were immersed in iodomethane solution for 7 days. NMR analysis of an 8.5 wt % ethanolic iodomethane solution showed that the concentration only decreased to 6.7 % within one week and 4.0 % within one month (Supporting Information, Fig. S2), suggesting that iodomethane in ethanol is stable enough to retain a high methylation capacity even after 1 month. Exclusion of light slows down discoloration of the iodomethane solution.

Based on the known structure of mTPBr and assuming quantitiave functionalizations, the IEC of the matrix polymer increases from 1.97 mmol OH^- g^{-1} for mTPBN/DABCO to 2.20 mmol OH^- g^{-1} for mTPN/TMA and 3.70 mmol OH^- g^{-1} for mTPN DABCO-Me. SEM analysis of the PPS fiber mat shows that the diameter of the PPS fibers is in the range of 10 μm and the thickness of the fiber mat is in the range of 90 μm. Analysis of the reinforced PPS/mTPBr membrane supports that both air and glass side are well pore-filled (Supporting Information, Fig. S3).

3.2. Membrane properties

Table 1 shows the water uptake and swelling of chloride exchanged membranes. Although mTPBN/TMA has a larger IEC than PPS/mTPN/DABCO, the water uptake is lower, presumably because the second amine group in DABCO increases the hydrophilicity of the membrane. Since alkaline stability is related with the concentration of hydroxide ions and their degree of solvation, this effect could result in a better alkaline stability than expected when comparing the alkaline stability of model compounds in KOH solution. Methylation of DABCO strongly increases the water uptake from 55 to 91 %.

While the thickness swelling nearly doubles from 9 % for the TMA membrane to 17 % for the DABCO-Me membrane, the length swelling is similar for all membranes, in the range of 9.8–11.4 %. This indicates that the PPS fibers limit the swelling in the area, which should reduce the mechanical stresses between assembly of a dry membrane into an electrolyser and start of operation.

The conductivity of the obtained membranes showed the expected trend and increased with increasing IEC and increasing concentration of the KOH solution (Fig. 2a). As reported before for several other membranes, the in-plane true hydroxide conductivity measured in DI water is higher than the through-plane conductivity in 0.5 M KOH, but lower than that in 1 M KOH [4]. This could be simply due to the different measurement methods or indicate anisotropic behavior, but since this behavior has been observed for several different membrane types, it seems plausible that the initially decreasing conductivity is related to the decreasing water contents in the membrane when the KOH concentration increases, which then is countered by uptake of excess

Table 1
Water uptake and swelling ratio of chloride exchanged PPS/mTPN/x membranes at room temperature.

Measurement	PPS/mTPN/TMA	PPS/mTPN/DABCO	PPS/mTPN/DABCO-Me
WU (%)	48.1 ± 8.8	54.9 ± 4.8	90.8 ± 5.2
SR_t (%)	9.0 ± 0.7	10.7 ± 0.6	17.1 ± 2.1
SR_L (%)	9.8 ± 3.1	11.2 ± 1.3	11.4 ± 0.2
SR_V (%)	31.5 ± 6.6	37.0 ± 3.8	45.3 ± 2.1

potassium and hydroxide ions when the KOH concentration increases further. It is noteworthy that the membrane with DABCO-Me groups has even higher conductivity in 2 M and 4 M KOH (ca. 10 and 19 wt% KOH, respectively) than Zirfon diaphragms (ca. 200 mS cm^{-1} in 20 wt% KOH) [24].

As Fig. 2b indicates, the membrane resistance decreases strongly when the KOH concentration increases from 0.5 M to 1 M, but levels off around 2 M. Therefore, in terms of membrane resistance, using membranes in 1 M KOH seems to be optimal to balance alkaline degradation and ASR. On the cell level, this could be different, because the ionic conductivity of the KOH feed solution influences the cell resistance.

Mechanical properties and the resistance against sharp edges under compressive forces are shown in supporting information Figs. S8 and S9, respectively.

3.3. Alkaline stability

Alkaline stability of AEMs is the major hurdle for widespread commercialization of AEMWE. Following the method used in previous work [17], the alkaline stability was tested in 1 M KOH at 60 °C. The degree of degradation was monitored by analysing the in-plane conductivity in a modifed Ziv-Dekel method, in which the conductivty cell containing the membrane sample is immersed in pure water, which is constantly deaerated by bubbling nitrogen. By applying a voltage, all carbonates are purged from the active membrane area, and the "true hydroxide conductivity" is measured [4,17].

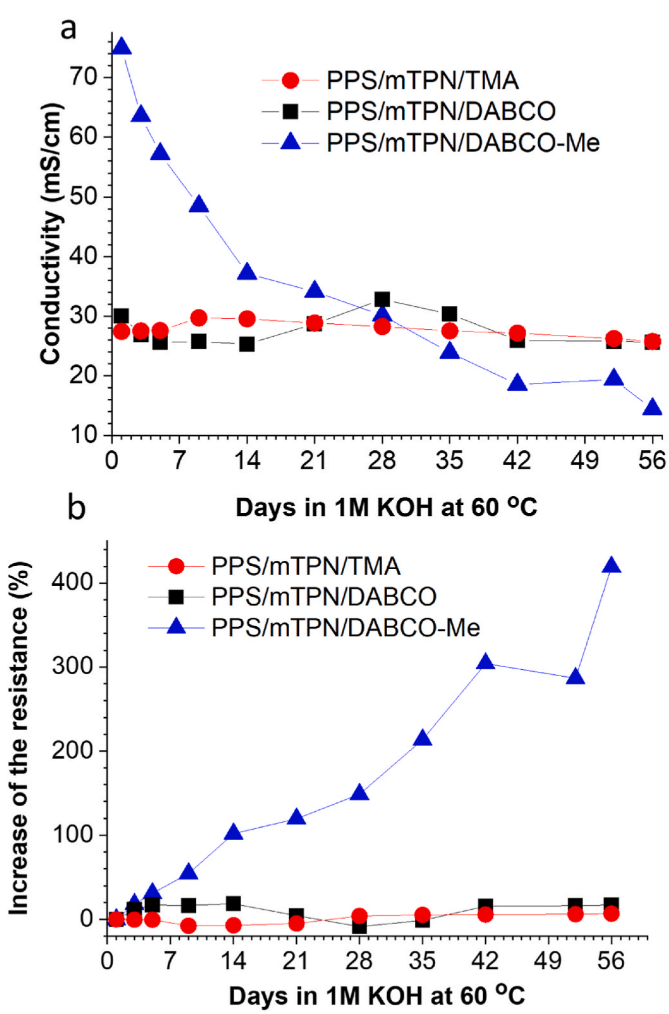

Fig. 3. Alkaline stability in 1 M KOH solution at 60 °C: (a) change of the in-plane conductivity, (b) change of the membrane resistance.

As shown in Fig. 3, the in-plane conductivity of PPS/mTPN/TMA remained practically unchanged over the full test duration of 56 days, as expected based on previous tests done with mTPN/TMA [4,21]. Bauer et al. analyzed the stability of model compounds, and suggested that DABCO-based membranes could be more stable than TMA-based membranes [25]. Although DABCO has β-hydrogen atoms, which potentially opens an alternative degradation pathway by Hofmann elimination, the C–H and C–N bonds of DABCO cannot be in the anti-periplanar orientation needed for the Hofmann elimination [16]. As a result, the alkaline stability of the PPS/mTPN/DABCO membrane was similar to that of the PPS/mTPN/TMA membrane.

The methylation of DABCO-modified polymers strongly increases the ionic conductivity, but early AEM literature indicated that membranes in which most DABCO moieties are used as a crosslinker, i.e. carrying two ammonium groups, are less stable than similar membranes with mono-functionalized DABCO groups [25]. The stability test shown in Fig. 3 confirms this. The initially ca. 2.5 times higher conductivity of PPS/mTPN/DABCO-Me strongly decreased and reached similar conductivity values as for TMA- and DABCO-modified PPS/mTPN within 28 days. While the resistance of PPS/mTPN/TMA and/DABCO increased less than 20 % over 56 days, that of PPS/mTPN/DABCO-Me increased over 400 %.

The reason why PPS/mTPN/DABCO-Me degrades faster than PPS/mTPN/DABCO is presumably that the high density of positive charges on DABCO-Me accelerates the initial reaction with hydroxide ions. Once one of the two ammonium groups is degraded by nucleophilic substitution at the CH_2 group, the remaining alkyl chain can freely rotate into a conformation which allows Hofmann elimination, which is sterically not possible in the rigid cage structure of DABCO. Thus, an additional degradation pathway is opened.

A surprisingly different result was obtained when stability was assessed by through-plane conductivity in 1 M KOH (Table 2). The through-plane conductivity of PPS/mTPN/DABCO-Me in this test was slightly higher than observed for the samples used for testing the conductivity in Fig. 2, probably caused by some batch-to-batch variation, but in the same range of ≈150 mS cm^{-1}. In contrast to the in-plane conductivity, which strongly decreased, the through-plane conductivity only decreased from 165 to 158 mS cm^{-1} within 30 days. Even more surprisingly, the conductivity of PPS/mTPN/DABCO increased from 46 to 200 mS cm^{-1}, which is close to the specific conductivity of 1 M KOH at 25 °C (215 mS cm^{-1}) [26].

A possible explanation for this is that some DABCO groups acted as dually quaternized crosslinker, which, as discussed before, is less alkaline stable than mono-quaternized DABCO. Such preferential removal of the crosslinking points would enlarge the hydrophilic domains and increase the absoprtion of KOH solution, which may have a higher conductivity than the external 1 M KOH solution due to the presence of hydroxide exchanged DABCO groups in the membrane. For example, a 4 M KOH solution has a conductivity of 570 mS cm^{-1} [26]. Since PPS/mTPN/DABCO-Me is synthesized from PPS/mTPN/DABCO, it can be assumed that the two membrane types have the same degree of crosslinking. Presumably, since DABCO has a relatively high alkaline stability, the IEC of the DABCO membrane does not much decrease between day 1 and day 30, while the degree of crosslinking decreases noticeably. The DABCO-Me membrane has a much higher initial IEC

Table 2
Alkaline stability in 1 M KOH solution at 60 °C: Change of the through-plane conductivity and ASR value.

Membranes	Conductivity (mS cm^{-1}) and ASR (in brackets, Ω cm^2)			
	0 day	1 day	30 days	6 months, then 6 months at RT
PPS/mTPN/DABCO	46.3 (216.2)	45.9 (235.1)	199.7 (58.1)	123.4 (93.2)
PPS/mTPN/DABCO-Me	165.3 (84.7)	164.9 (89.7)	158.0 (78.5)	70.3 (186.5)

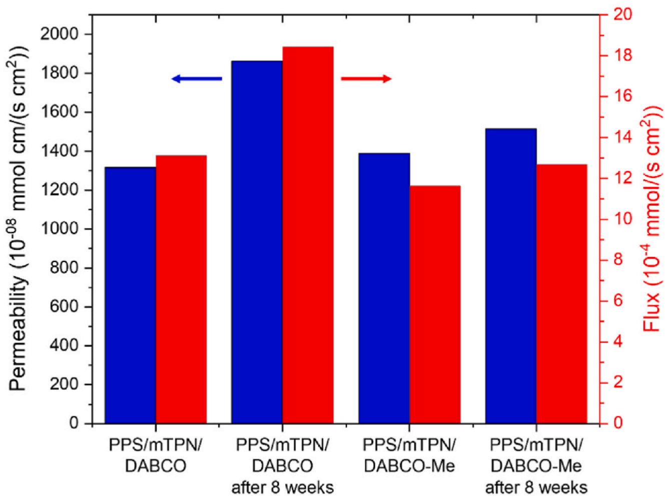

Fig. 4. Vapor/vapor permeation of PPS/mTPN membranes before and after alkaline stability test shown in Fig. 3.

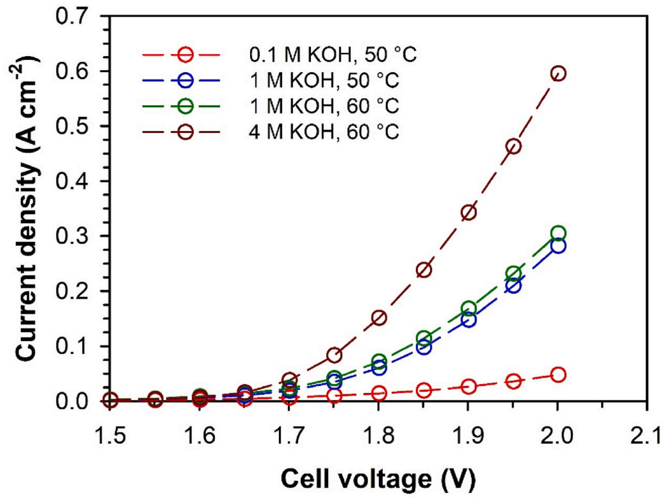

Fig. 5. Load curves of an alkaline water electrolyser under different KOH concentrations and temperatures. PPS/mTPN/DABCO-Me as solid electrolyte and substrate for catalyst layers, PSEBS-CM-DABCO as polymer binder, cathode catalyst – H2GEN-M, anode catalyst – OXYGN-N, catalyst load 2.5 mg cm^{-2}, catalyst to binder ratio 93:7, Ni foam as porous transport layer. Temperature and KOH concentration are shown in the figure.

than the DABCO membrane, and may absorb also more additional KOH than the DABCO membrane, which results in a higher conductivity at day 1. Over time, similar to the DABCO membrane, degradation reduces the degree of crosslinking, which results in larger absorption of water per ionic group, but simultanouelsy, also the IEC decreases, since the DABCO-Me groups are less stable than DABCO groups. In combination, these two effects result in a nearly unchanged through-plane conductivity for the DABCO-Me membrane.

Enlarged hydrophilic domains should also result in increased water transport over the membrane. Indeed, the water vapor/vapor permeation through a PPS/mTPN/DABCO membrane increased 40 % from 13.1 to 18.4 mmol H_2O s^{-1} cm^{-2} over the test time of 30 days (Fig. 4), supporting that the volume of the hydrophilic domains increased.

While through-plane conductivty was measured in KOH solution, in-plane conductivty was measured in pure water. Therefore, enlarged hydrophilic domains will have a positive effect on the conductivty by reducing the tortuosity, but a negative effect by reducing the concentration of mobile hydroxide ions, which could explain the stable conductivity observed in Fig. 3.

In conclusion, the alkaline stability tests suggest that loss of quaternary ammonium groups strongly increases the resistance of PPA/mTPN/DABCO-Me in pure water, but the excess KOH absorbed into the hydrophilic domains retains high conductivity also when some of the quaternary ammonium groups are degraded, shifting the membrane properties from AEM to a hybrid of AEM and ISM (ion solvating membrane, Fig. S12 explains the function of AEM, ISM and diaphragms.) Therefore, it is expected that PPS/mTPN/DABCO and PPS/mTPN/DABCO-Me can function well as separator in AEMWE operated with an alkaline feed solution also over a longer time duration. Indeed, through-plane conductivity of DABCO and DABCO-Me membranes after storage for 6 months in 1 M KOH at 60 °C, followed by 6 months at RT, still was 123 and 70 mS cm^{-1}, respectively. However, the membranes are expected to soften over time and may show an increased gas crossover, because gas crossover is expected to occur mainly by transport of dissolved gases through the hydrophilic domains.

3.4. Alkaline membrane water electrolyser performance

Load curves of an alkaline water electrolyser using the PPS/mTPN/DABCO-Me membrane under different operational conditions are shown in Fig. 5. A strong influence of the KOH concentration can be seen from comparison of the load curves measured at 50 °C and KOH concentrations of 0.1 and 1 M KOH. It is obvious that using 1 M KOH performance increased dramatically when compared to the performance achieved in 0.1 M KOH. The results of EIS (Fig. 6, samples 1 and 2) shows that the performance improvement with higher KOH concentration is due to both Rs and Rp values decrease. With respect to the cell setup, the main contribution to the Rs value comes from the MEA. Reason for higher ionic conductivity of the PPS/mTPN/DABCO-Me membrane can lie in weaker Donnan exclusion due to higher ionic strength of the feed solution. Weaker Donnan exclusion would allow more K^+ ions to penetrate into the anion-selective phase of the PPS/mTPN/DABCO-Me membrane increasing thus, due to electroneutrality preservation, number of mobile OH^- ions. Another explanation lies in the ion solvation, when KOH electrolyte penetrates into the bulk of the membrane. Both effects are connected with higher ionic conductivity of the membrane and consequently lower Rs value. Lower Rp values are mainly achieved due to the improved ionic contact in the catalyst layer, which increases the number of the catalyst particles available for electrochemical reaction.

Except for KOH concentration influence on performance of the alkaline water electrolysis, the effect of the temperature is shown in Fig. 5 too. Increase of the temperature from 50 to 60 °C (1 M KOH) resulted in further improvement of the alkaline water electrolysis performance, when current density at cell voltage 2.0 V increased by 7.8 %. Based on the EIS data shown in Fig. 6 (samples 2 and 3), achieved improvement can be mainly addressed to the decrease in Rp values (22.7 %) but Rs value decrease was observed too (12.9 %). This observation is not surprising, as higher temperature accelerates the reaction rate and thus reduces the polarization resistance and increases the ionic conductivity of the liquid and polymer electrolyte.

Based on results of the concentration and temperature influence measurements, experiment with 4 M KOH at 60 °C has been performed. It is clearly seen form Fig. 5 that further increase in KOH concentration from 1 to 4 M KOH improved the performance of alkaline water electrolysis even more. At the cell voltage of 2.0 V the achieved current density increased from 0.305 to 0.595 A cm^{-2} (increase by 95.4 %), compared to that with 1 M KOH feed at 60 °C. EIS confirmed that the improvement is once again due to both, Rs and Rp values decrease (Fig. 6).

It can thus be concluded that using KOH electrolyte of high concentration has a positive impact on the alkaline water electrolysis performance using PPS/mTPN/DABCO-Me membrane. This effect is achieved by improved ionic conductivity of the polymer materials,

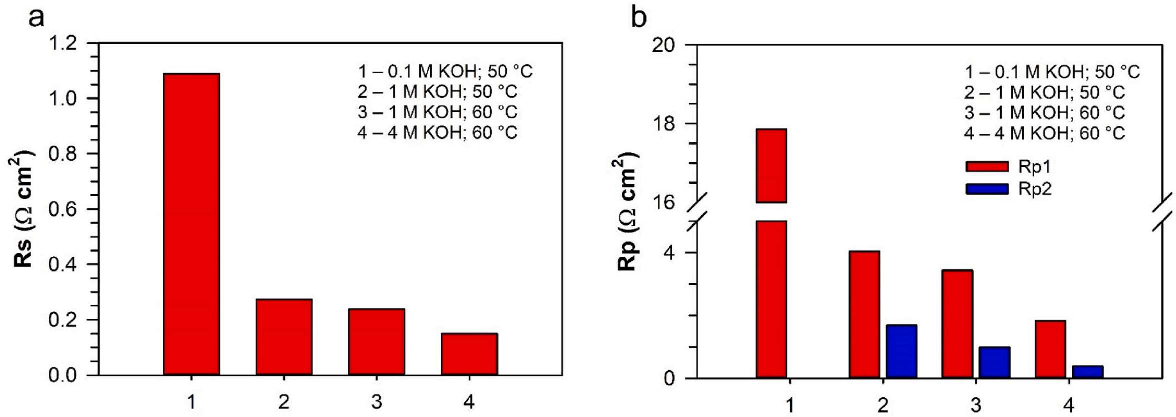

Fig. 6. EIS data of the Rs and Rp values of the alkaline water electrolysis under different KOH concentrations and temperatures. EIS measured at 1.5 V (Rs) and 1.7 V (Rp), frequency range 20 kHz–0.1 Hz, maximal amplitude of perturbing signal 20 mV, PPS/mTPN/DABCO-Me as solid electrolyte and substrate for catalyst layers, PSEBS-CM-DABCO as polymer binder, cathode catalyst – H2GEN-M, anode catalyst – OXYGN-N, catalyst load 2.5 mg cm^{-2}, catalyst to binder ratio 93:7, Ni foam as porous transport layer.

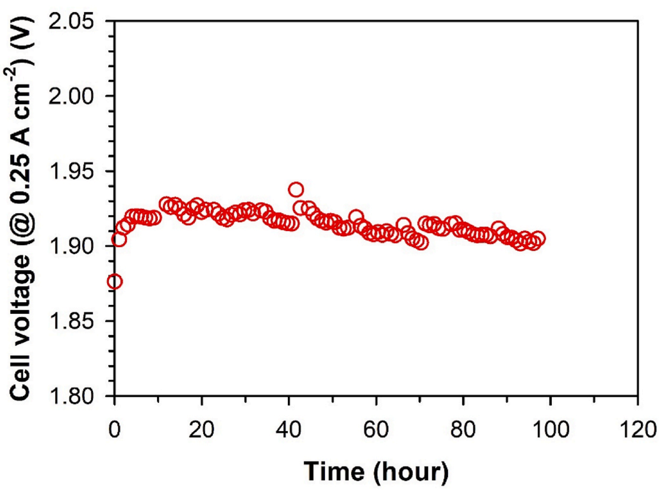

Fig. 7. Stability measurement of PPS/mTPN/DABCO-Me. 4 M KOH, 60 °C, 0.25 A cm^{-2}, PPS/mTPN/DABCO-Me as solid electrolyte and substrate for catalyst layers, PSEBS-CM-DABCO as polymer binder, cathode catalyst – H2GEN-M, anode catalyst – OXYGN-N, catalyst load 2.5 mg cm^{-2}, catalyst to binder ratio 93:7, Ni foam as porous transport layer.

higher utilization of the catalyst in the catalyst layer and improved kinetics of the electrode reactions.

However, besides the performance, the stability of the MEA represents important factor. In order to evaluate this parameter for the PPS/mTPN/DABCO-Me membrane, the stability test was performed at the harshest conditions, i.e. 4 M KOH and 60 °C. Fig. 7 shows cell voltage values at constant current density of 0.25 A cm^{-2} within 100 h test. Fig. 7 shows increase of the cell voltage from initial 1.88 V–1.92 V in the first 10 h of the stability experiment. This period is however followed by the cell voltage stabilization. Deterioration of alkaline water performance and its stabilization are confirmed by the load curve measurement (Fig. 8a). EIS results (Fig. 8b) show the increase of the Rs value after first 20 h of stability test by 22.4 %. This is in agreement with observed ionic conductivity decrease showed in Fig. 3. However, the value of ionic conductivity is gradually decreasing in Fig. 3a in contrary to the observed stabilization of Rs value in Fig. 8b. Explanation of this contradiction lies in the effect of the KOH solution as shown in Table 2. Initial increase of the Rs can thus be attributed to the functional groups degradation, besides the stabilization is achieved due to the presence of the KOH in the bulk of the PPS/mTPN/DABCO-Me membrane.

With respect to the Rp values, Fig. 8b shows the fluctuation of the Rp1 value, which is the highest from all the obtained resistance values and gradual increase of the Rp2 value. This increase can be due to the catalyst degradation or loss.

4. Conclusions

A series of AEMs was fabricated, in which all membranes have an ether-free poly(arylene) backbone based on mTPN, are reinforced with a PPS fiber mat to control in-plane swelling, but have different quaternary ammonium groups. By changing DABCO group to TMA and methylated DABCO, the IEC can be adjusted between 1.97 and 3.70 mmol OH^- g^{-1}.

DABCO–membranes have lower IEC than TMA-functionalized membranes, but a higher water uptake of 55 %, presumably because the amine group increases the hydrophilicity. While methylation of DABCO increases the water uptake to 91 %, the in-plane swelling is limited by the PPS reinforcement, and increases only from 11.2 to 11.4 %.

Changing the KOH concentration from 0.5 M to 1 M KOH strongly reduces the membrane ASR, but changing from 2 M to 4 M KOH only shows a small effect. This indicates that 1M–2M KOH is sufficient to achieve optimal membrane ASR without risking alkaline degradation of components. To prove this, membranes were immersed in 1 M KOH at 60 °C for 2 months. It was found that PPS/mTPN/DABCO-Me rapidly lost in-plane conductivity, but that PPS/mTPN/TMA and PPS/mTPN/DABCO were practically unchanged. However, while conductivity and ASR did not change much, a small increase in the water vapor/vapor permeation was observed, which could indicate pinhole formation or an increased dimension or connectivity of the hydrophilic domains.

In the electrolyser, performance increased with increasing KOH concentration. Interestingly, polarization resistance contributed more to overall resistance than ohmic resistance. Over 100 h test, the PPS/mTPN/DABCO-Me membrane remained stable, different than predicted by the ex-situ stability test using in-plane conductivity in DI Water. It seems that the membrane lost some quaternary ammonium groups, but still kept the absorbed KOH solution, which suggests that the membrane did not function as a perfect AEM, but as a combination of AEM and ion solvating membrane (ISM). This finding could open a strategy towards enhanced membrane life-time, by adding hydrophilic groups to the membrane, which can help to bind excess KOH and water even after the ion-selective groups degrade.

CRediT authorship contribution statement

Malikah Najibah: Writing – review & editing, Writing – original

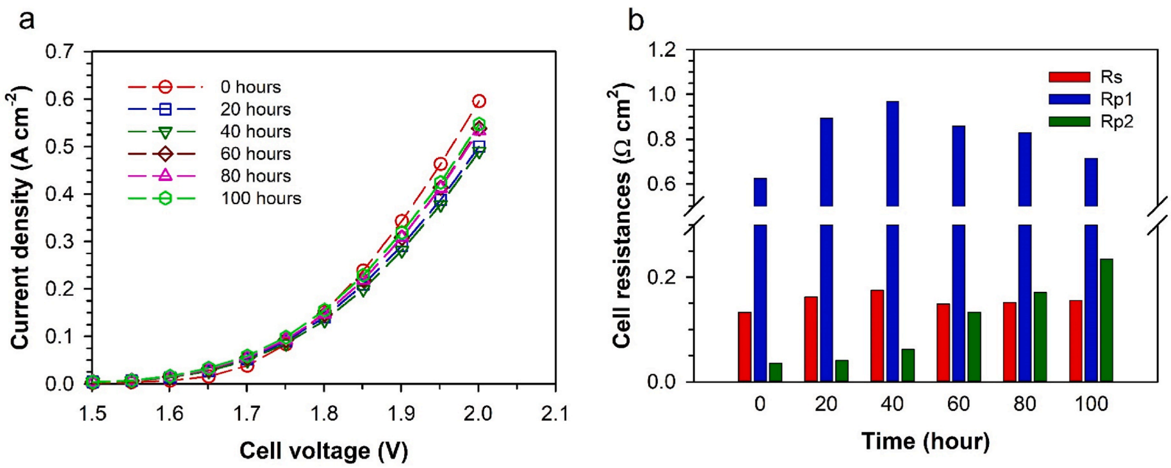

Fig. 8. Load curves and EIS analysis of the stability test. 4 M KOH, 60 °C, EIS measured at 1.5 V (Rs) and 1.7 V (Rp), frequency range 20 kHz–0.1 Hz, maximal amplitude of perturbing signal 20 mV, PPS/mTPN/DABCO-Me as solid electrolyte and substrate for catalyst layers, PSEBS-CM-DABCO as polymer binder, cathode catalyst – H2GEN-M, anode catalyst – OXYGN-N, catalyst load 2.5 mg cm^{-2}, catalyst to binder ratio 93:7, Ni foam as porous transport layer.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2024.123335.

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